



Stellar Classification Using Machine Learning: A Comparative
Analysis of Supervised and Unsupervised Learning Approaches
with SMOTE Optimization.

MACHINE LEARNING – CMP7228

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Abstract

This report presents an in-depth analysis of stellar classification using machine learning techniques on a strategically sampled subset of the Sloan Digital Sky Survey (SDSS) dataset. The analysis implements three machine learning approaches: Random Forest Classifier and Extra Trees Classifier for supervised learning, and K-means clustering for unsupervised learning. Using stratified sampling to ensure representative class distribution, 10,000 stellar observations were analyzed through systematic data preprocessing, feature engineering, and model evaluation. The analysis focuses on handling high-dimensional spectral data and addressing class imbalance challenges inherent in astronomical datasets. Comprehensive evaluations, supported by visualizations and statistical analyses, demonstrate the practical applicability of these machine learning techniques in astronomical classification, providing insights for future applications in this domain.

1. Domain Description

1.1 Introduction to Stellar Classification

Stellar classification represents a cornerstone of astronomical research, serving as "the principal key to our understanding of stellar evolution and the structure of the universe" (York et al., 2000). This systematic categorization of celestial objects, primarily based on their spectral characteristics, has evolved from early visual observations to sophisticated spectroscopic analysis. The integration of machine learning techniques with traditional classification methods has revolutionized our approach to analyzing astronomical data (Zhou and Zhu, 2019).

1.2 The Modern Classification System

The Morgan-Keenan (MK) classification system provides a comprehensive framework for stellar categorization through a multi-dimensional approach:

- Spectral temperature sequence (O, B, A, F, G, K, M)
- Luminosity classes (I through V)
- Chemical composition indicators
- Peculiarity notations

This standardized approach correlates observable spectral features with underlying physical properties, establishing what Wang et al. (2020) term "a bridge between observation and theoretical astrophysics."

1.3 Importance in Modern Astronomy

Stellar classification is crucial for understanding **stellar evolution**, **galactic structures**, and the **discovery of rare stellar objects** like neutron stars and black holes (York et al., 2000; Zhou & Zhu, 2019). It also aids in the **search for exoplanets** by identifying stars with Sun-like properties, narrowing the focus on potentially habitable systems (Zhang, 2019; Wang & Zheng, 2021).

1.4 Role of Machine Learning

Machine learning models can analyze large astronomical datasets to identify patterns and classify stars with unprecedented accuracy and speed (Zhou and Zhu, 2019). The exponential growth in data from sky surveys like SDSS has necessitated automated classification methods. These automated systems not only reduce classification time but also improve accuracy and consistency (Zhou Zhihua, 2016).

The combination of traditional astronomical knowledge with modern machine learning techniques represents a significant advancement in our ability to understand and categorize celestial objects, enabling discoveries that would be impossible through manual analysis alone (Zhang, 2019).

2. Problem Statement

The exponential growth of astronomical data from large-scale sky surveys, particularly the Sloan Digital Sky Survey (SDSS), presents a significant challenge in efficient classification of celestial objects (York et al., 2000). Traditional manual classification processes, while foundational, cannot effectively manage the volume and complexity of modern astronomical data (Wang et al., 2021). This limitation creates a pressing need for automated, accurate, and scalable methods of stellar classification (Zhou Zhihua, 2016).

Machine learning offers a promising solution by leveraging algorithms that can learn patterns from data, automating classification processes while potentially improving accuracy (Zhou and Zhu, 2019). However, implementing these models presents specific challenges in algorithm selection, data preprocessing, and performance evaluation.

This study aims to develop and evaluate machine learning models for accurate stellar classification using SDSS spectral data, with specific objectives to:

1. Implement and compare multiple machine learning approaches for stellar classification (Breiman, 2001)
2. Evaluate preprocessing techniques' effectiveness on classification accuracy (Cristianini and Shawe-Taylor, 2000)

3. Analyze feature selection impact on model performance (Zhou Zhihua, 2016)
4. Develop a robust classification system for large astronomical datasets (Zhang, 2019)
5. Identify reliable indicators for stellar classification (York et al., 2000)

The key challenges addressed include managing high-dimensional spectral data while maintaining classification accuracy across stellar types, and addressing class imbalance issues common in astronomical datasets (Wang et al., 2021). Through the implementation of machine learning techniques, this project contributes to ongoing efforts in automating stellar classification and advancing our understanding of the cosmos.

3. Dataset Description

3.1 Data Source and Sampling

The dataset originates from the Sloan Digital Sky Survey (SDSS) Data Release 17, accessed through Kaggle (Fedesoriano, 2022). From the original 100,000 observations, a strategically sampled subset of 10,000 observations was selected using stratified sampling techniques to maintain class distribution while reducing computational complexity.

	Original Distribution (%)	Reduced Distribution (%)
class		
GALAXY	59.445	59.45
STAR	21.594	21.59
QSO	18.961	18.96

Figure 1: Comparison of Class Distributions Before and After Stratified Sampling

3.2 Dataset Characteristics

The sampled dataset comprises:

- 10,000 observations
- 17 features (excluding target variable)
- Target Variable: Class (Galaxy, Star, or Quasar)
- Class Distribution: GALAXY (59.5%), STAR (21.6%), QSO (19.0%)

3.3 Feature Description

The dataset includes three primary feature categories:

1. Object Identity and Position

- obj_ID: Unique object identifier
 - alpha: Right Ascension angle (J2000 epoch)
 - delta: Declination angle (J2000 epoch)
2. Photometric Parameters
- u: Ultraviolet filter (3551Å) magnitude
 - g: Green filter (4686Å) magnitude
 - r: Red filter (6165Å) magnitude
 - i: Near Infrared filter (7481Å) magnitude
 - z: Infrared filter (8931Å) magnitude
3. Spectroscopic and Measurement Parameters
- redshift: Light wavelength shift measurement
 - spec_obj_ID: Spectroscopic measurement identifier
 - run_ID, rerun_ID: Imaging identifiers
 - cam_col: Camera column identifier
 - field_ID: Field of view identifier
 - plate: Plate ID
 - MJD: Modified Julian Date
 - fiber_ID: Fiber identifier

3.4 Target Classes

GALAXY:

A galaxy represents a gravitationally bound system of stars, stellar remnants, interstellar gas, dust, and dark matter. Galaxies contain an estimated 100 million stars on average, ranging from dwarf galaxies with fewer than a hundred million stars to supergiants containing up to one hundred trillion stars. Most of their mass exists as dark matter, with only a small percentage visible as stars and nebulae.

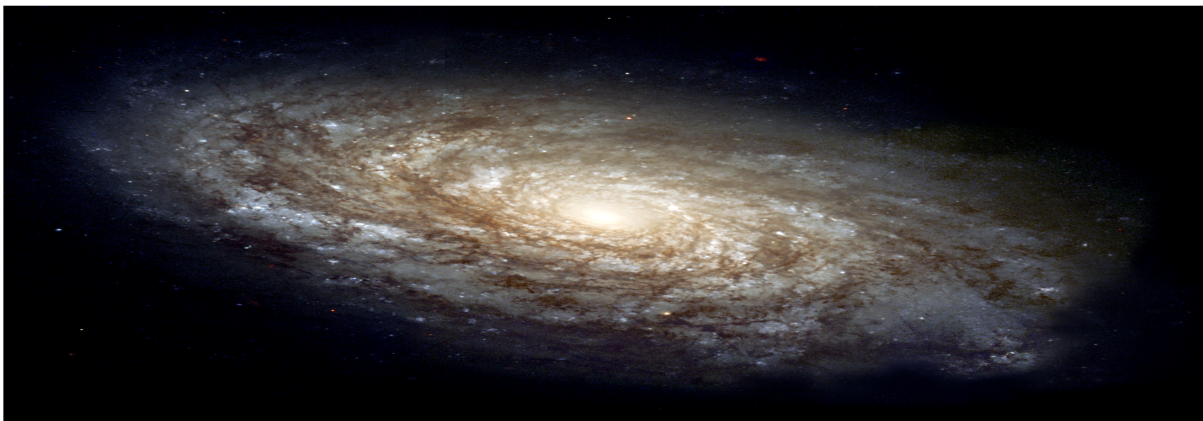


Figure 2: A Spiral Galaxy in Space

STAR :

Stars begin as gravitational collapses of gaseous nebulae, primarily composed of hydrogen and helium. Their mass primarily determines their evolution and fate. Stars generate energy through thermonuclear fusion of hydrogen into helium in their cores. At life's end, they become stellar remnants: white dwarfs, neutron stars, or black holes, depending on their initial mass.



Figure 3: Stars in the Cloud

QSO (Quasar):

Quasars are extremely luminous active galactic nuclei (AGN), powered by supermassive black holes ranging from millions to tens of billions of solar masses. These objects are surrounded by gaseous accretion discs where infalling matter generates intense electromagnetic radiation. Quasars possess luminosities thousands of times greater than typical galaxies like the Milky Way.



Figure 4: Quarsar in space

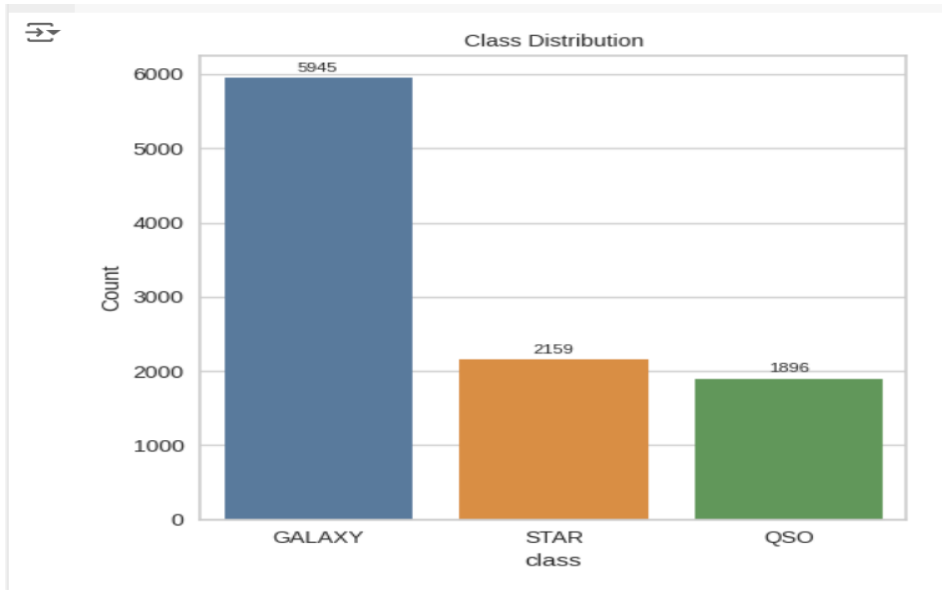


Figure 5: Bar chart showing Class Distribution of Stellar Dataset

These characteristics influenced the preprocessing strategy and motivated the use of stratified sampling to create a more manageable yet representative dataset for analysis.

3.5 Data Quality Assessment

Key characteristics of the dataset include:

- Significant class imbalance (GALAXY: 5,945, STAR: 2,159, QSO: 1,896)
- Professional-grade measurements from SDSS instrumentation
- Complete data with no missing values
- Raw spectroscopic and photometric measurements requiring preprocessing

3.6 Dataset Limitations

Notable constraints include:

- Limited to bright objects detectable by SDSS
- Northern hemisphere visibility restriction
- Potential spectroscopic selection effects
- Survey magnitude limits affecting completeness

4. Exploratory Data Analysis

The analysis follows three key levels: univariate, bivariate, and multivariate analysis, revealing crucial patterns for stellar classification (Zhang, 2019).

4.1 Univariate Analysis

Class distribution analysis revealed inherent class imbalance typical in astronomical surveys (York et al., 2000):

- Galaxies: 59.5%
- Stars: 21.6%
- Quasars: 19.0%

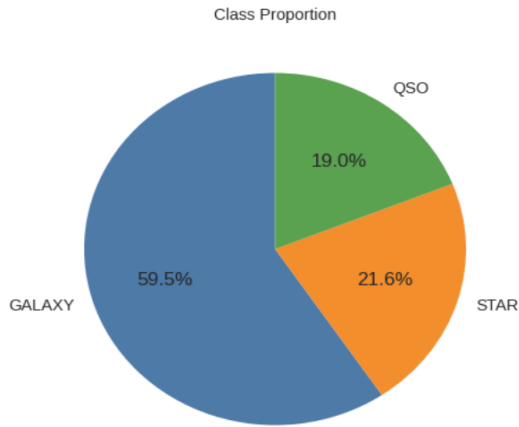


Figure 6: Pie Chart Class Distribution

Statistical analysis exposed distinctive feature characteristics:

- Most features exhibit slight negative skewness
- Redshift shows significant positive skew (2.62)
- Alpha and delta features: kurtosis values of -0.54 and -1.03
- Redshift demonstrates high kurtosis (10.69)

	alpha	delta	u	g	r	i	z	redshift
Mean	176.4133	24.0728	22.0578	20.6128	19.6287	19.0657	18.7549	0.5817
Median	180.4910	23.4565	22.1790	21.0607	20.0867	19.3706	18.9730	0.4122
Mode	0.0448	-17.5738	24.6346	17.0976	16.0129	15.8764	22.8269	0.0000
Std Dev	96.7562	19.6676	2.2420	2.0396	1.8629	1.7609	1.7782	0.7498
Range	359.9488	100.5743	15.3785	15.6695	18.7081	21.5750	18.0117	7.0169
IQR	105.6280	34.6565	3.3635	3.1859	2.9632	2.6773	2.4646	0.6415
Variance	9361.7571	386.8138	5.0264	4.1599	3.4705	3.1007	3.1620	0.5623
Skewness	-0.0275	0.1800	-0.0809	-0.4148	-0.4453	-0.3436	-0.1644	2.6176
Kurtosis	-0.5436	-1.0326	-0.5673	-0.4302	-0.3577	-0.0923	-0.0900	10.6859
Min	0.0448	-17.5738	13.9472	11.7452	10.8638	10.5665	10.7789	-0.0057
Max	359.9936	83.0005	29.3256	27.4146	29.5719	32.1415	28.7906	7.0112
25%	126.7633	5.2548	20.3237	18.9465	18.0766	17.6900	17.4254	0.0561
75%	232.3913	39.9113	23.6872	22.1324	21.0398	20.3673	19.8900	0.6976

Figure 7: Statistical Summary of Numerical Features

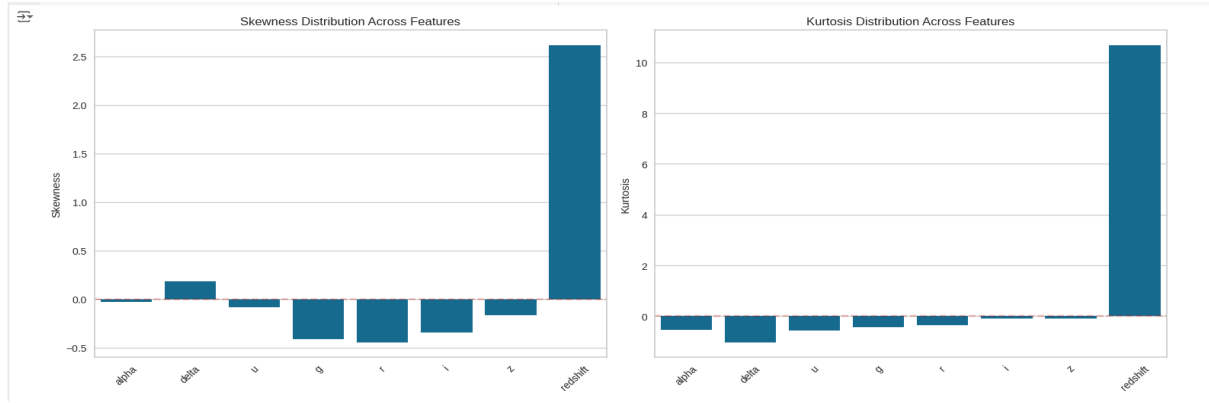


Figure 8: Bar charts showing skewness and kurtosis distributions across features

Photometric bands (u, g, r, i, z) analysis revealed:

- U-band (ultraviolet): Mean 22.06, SD 2.24, right-skewed
- G-band (green): Mean 20.61, SD 2.04, bimodal distribution
- R-band (red): Mean 19.63, SD 1.86, mirrors g-band bimodality
- I-band and Z-band: Symmetrical distributions (means 19.07, 18.75)

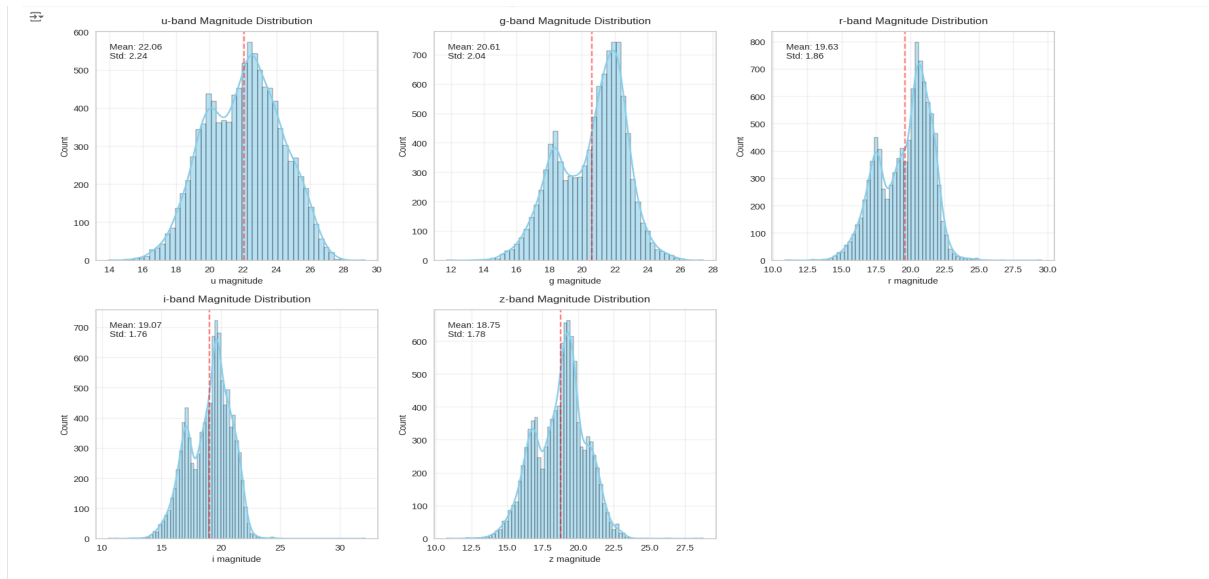


Figure 10: Photometric Band Magnitude Distributions

Kernel Density Estimation showed distinct class-specific patterns (Wang & Zheng, 2021):

- Galaxies: Broader distributions with lower magnitudes
- Quasars: Characteristic peaks in u and g bands
- Stars: Uniform distributions across bands

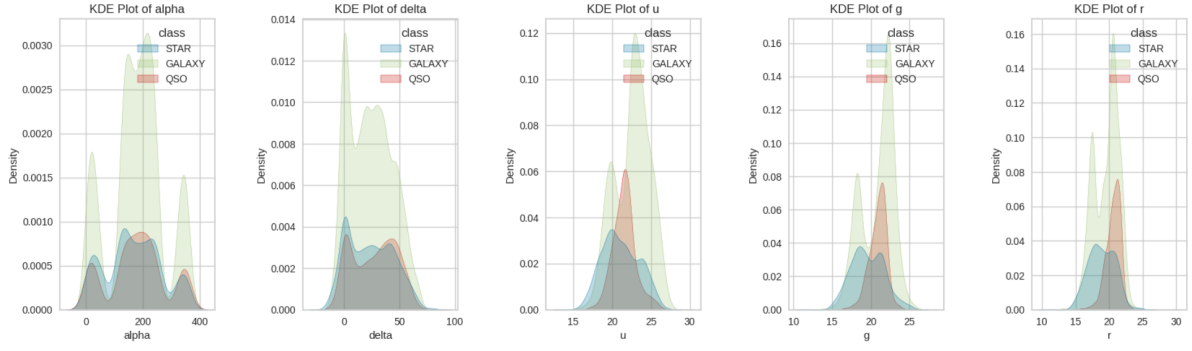


Figure 11a: KDE Plots for features

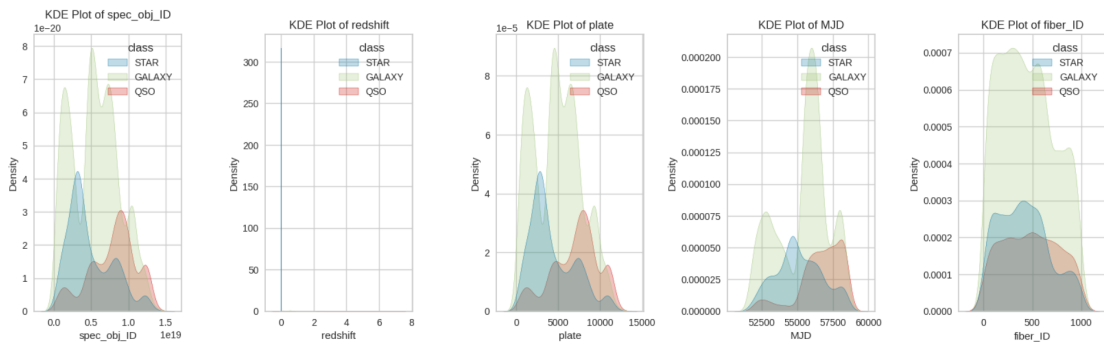


Figure 11b: KDE Plots for features

KDE plot analysis highlights key patterns. The spatial features, **alpha** and **delta**, reveal galaxy clustering aligned with large-scale structures (York et al., 2000). Photometric bands show **galaxies** with broader distributions and lower magnitudes, **quasars** with peaks in **u** and **g** bands indicating high-energy emissions, and **stars** with uniform distributions across bands (Zhou & Zhu, 2019).

Refer to **Appendix D** to view more **Univariate Plots** like Skewness and Kurtosis Feature Distributions, Box Plots Feature Distributions, and Violin Plots Feature Distributions.(Figure 9, Figure 12a, Figure 12b, Figure 13a, Figure 13b)

4.2. Bivariate Analysis

Redshift patterns revealed clear class separation:

- Stars: Minimal values (Milky Way proximity)
- Galaxies: Moderate spread
- Quasars: Broad, heavy-tailed distributions

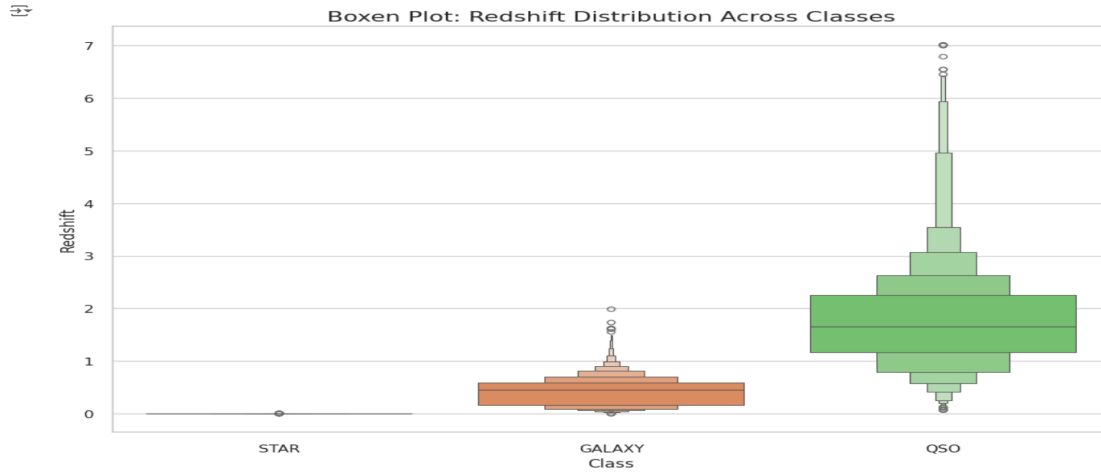


Figure 14: Boxen Plot of Redshift Distribution

4.3. Multivariate Analysis

Position and distance features showed minimal correlation (max ~ 0.15), indicating independent spatial information (York et al., 2000).

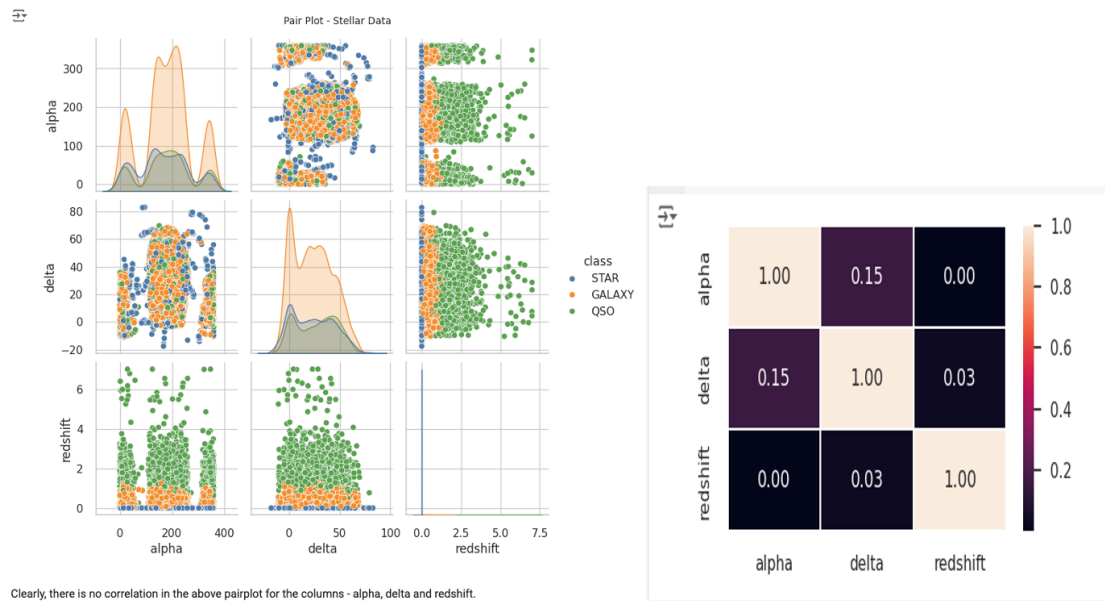


Figure 15: Pairplot and Correlation Matrix For alpha, delta, and redshift

Photometric bands demonstrated strong correlations:

- Highest: i-band and z-band (0.97)
- Lowest: u-band and z-band (0.54)

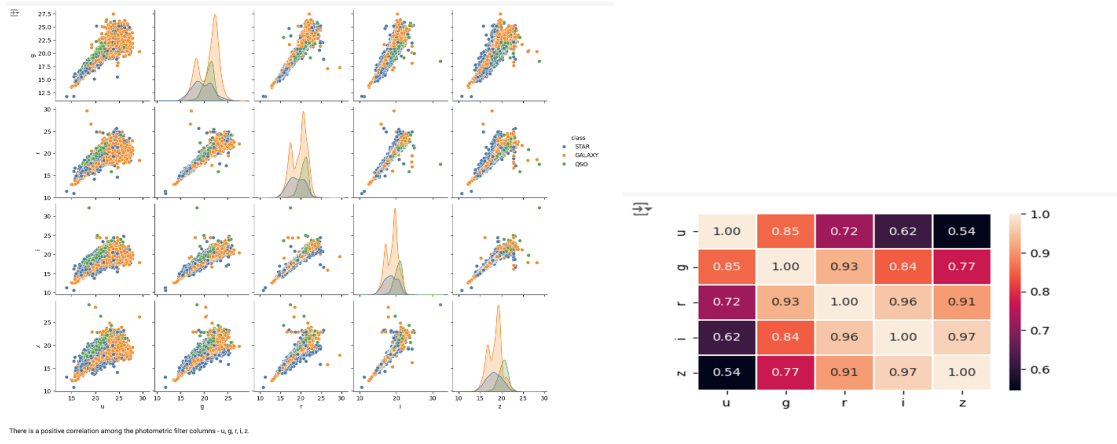


Figure 18: Pairplot and Correlation Matrix for Photometric Filter Columns - u, g, r, i, z

Three-dimensional analysis of Right Ascension, Declination, and redshift revealed distinct clustering by class (Zhou & Zhu, 2019).

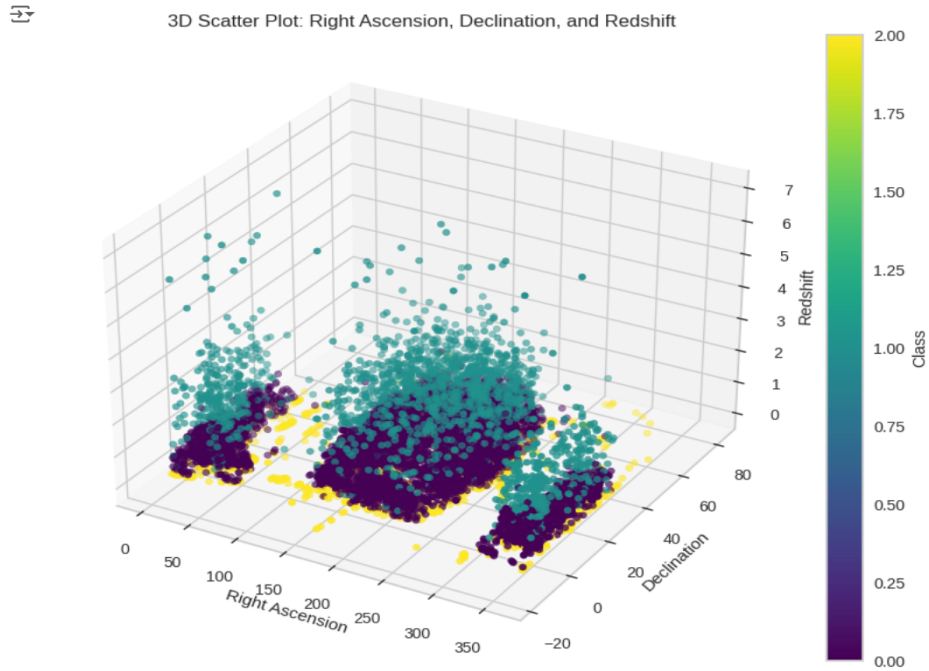


Figure 19: 3D Scatter Plot: Right Ascension, Declination and Redshift

5. Data Preprocessing

5.1. Feature Engineering and Selection

Initial correlation analysis using Pearson and Spearman heatmaps revealed strong relationships between photometric bands (u, g, r, i, z) with correlations ranging from 0.54 to 0.97 (Wang & Zheng, 2021). These relationships informed our feature selection strategy.

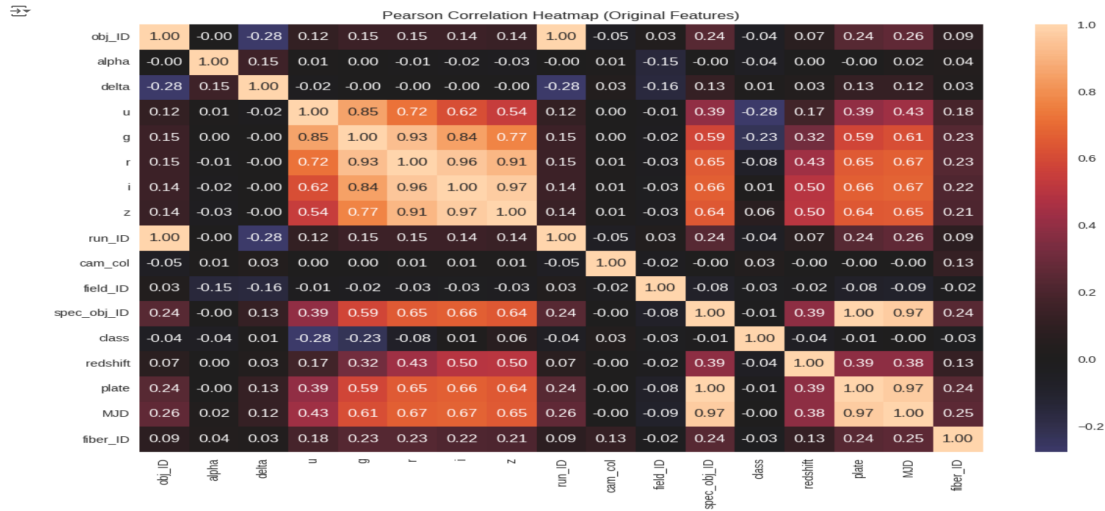


Figure 20: Correlation Analysis Before Feature Engineering - Pearson Heatmap

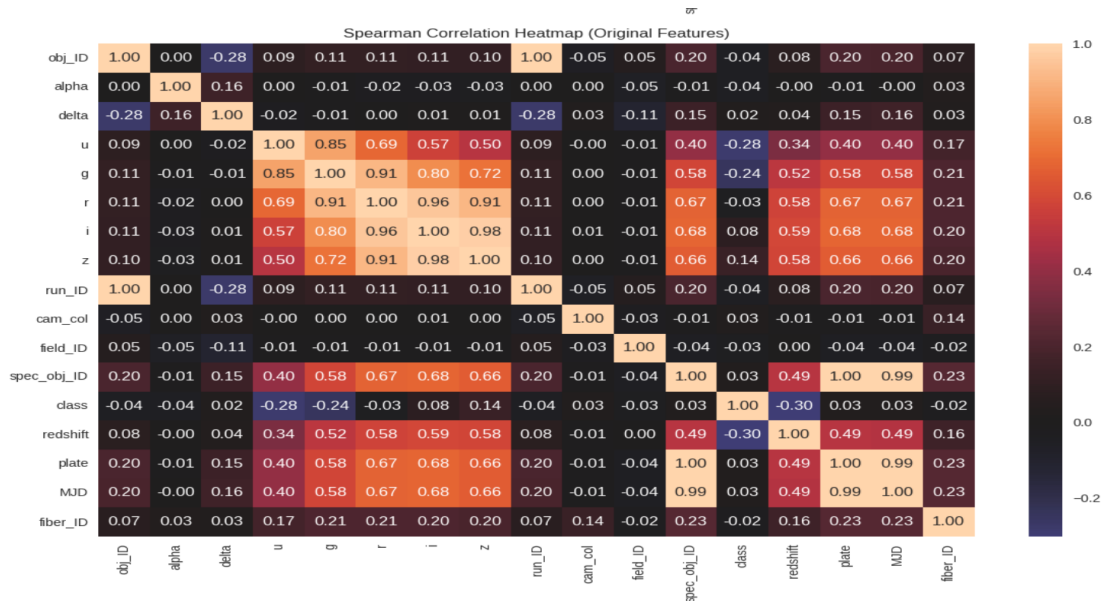


Figure 21: Correlation Analysis Before Feature Engineering - Spearman Heatmap

Principal Component Analysis (PCA) guided feature selection by revealing component contributions (Zhang, 2019):

- First component: Strong photometric feature loadings (-0.27 to -0.37)
- Second component: Dominated by identification features, particularly obj_ID and run_ID (0.64)

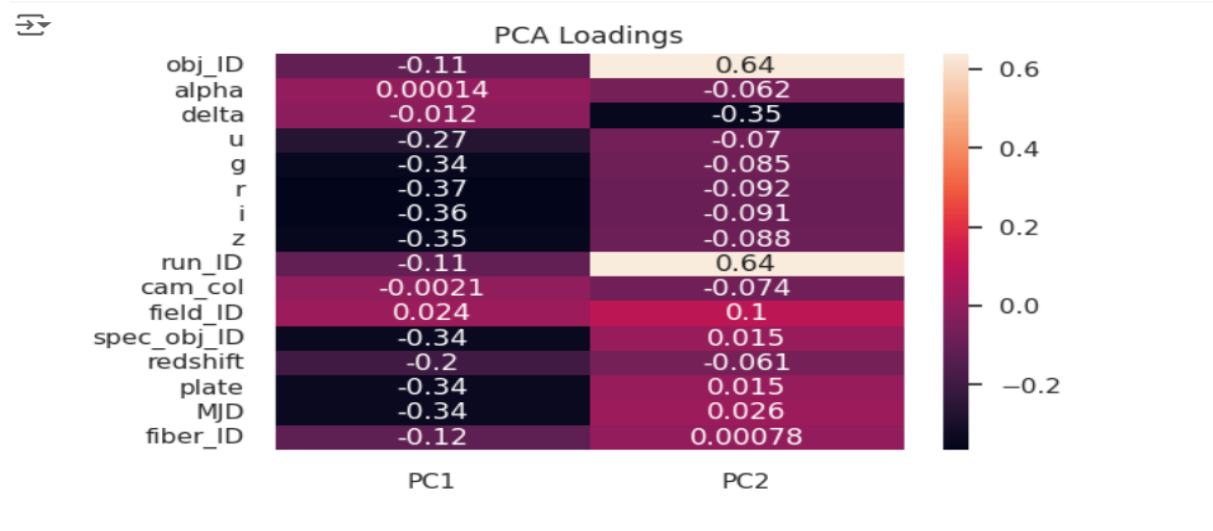


Figure 22: PCA Loadings

Based on PCA results, we selected features: ['u', 'g', 'r', 'i', 'z', 'spec_obj_ID', 'redshift', 'plate', 'MJD']. Random Forest feature importance analysis identified redshift as the most significant predictor (importance score ~0.6), followed by photometric bands.

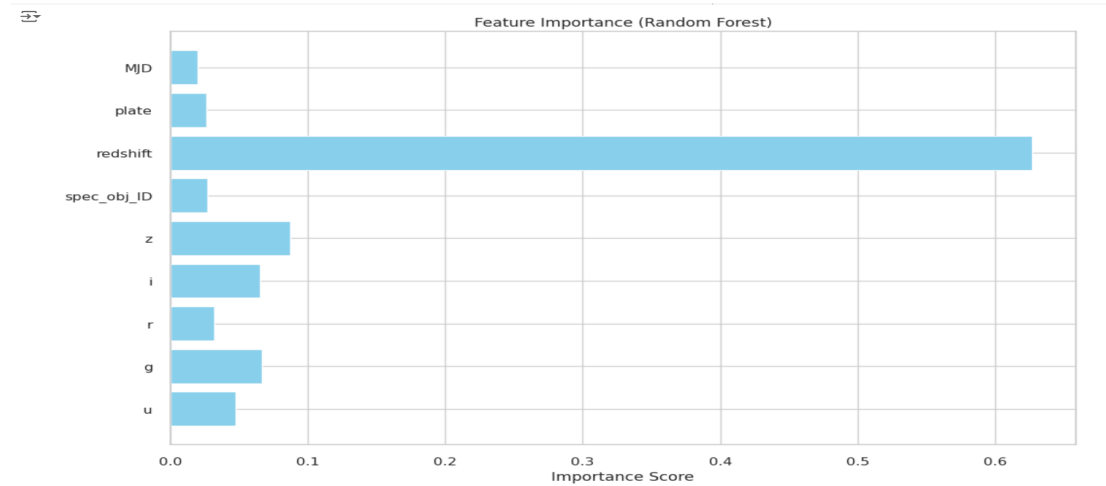


Figure 23: Random Forest feature importance plot

Statistical validation through Mutual Information and ANOVA F-Test scores confirmed feature relevance, with the ANOVA F-test showing significant between-group variance for chosen features (Zhou & Zhu, 2019).

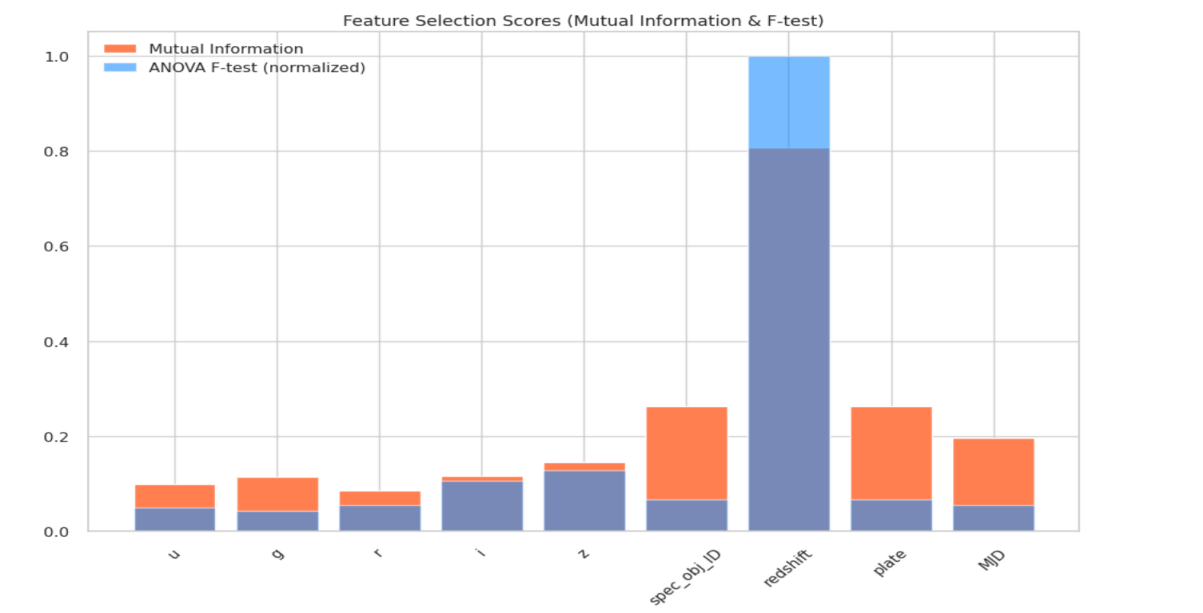


Figure 24: Mutual Information and ANOVA F-Test scores

Post-feature engineering correlation analysis (Figures 25 and 26) confirmed the effectiveness of our feature selection process, revealing optimized relationships between selected features while reducing redundancy (Wang & Zheng, 2021).

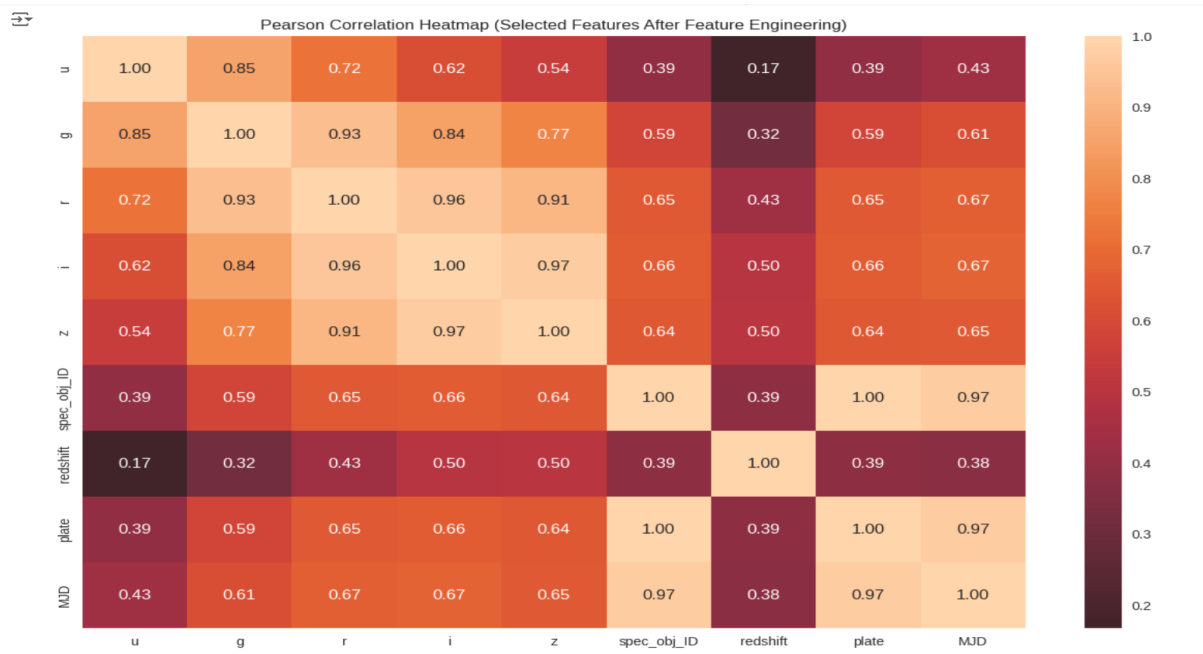


Figure 25: Correlation Analysis After Feature Engineering - Pearsonn Heatmap

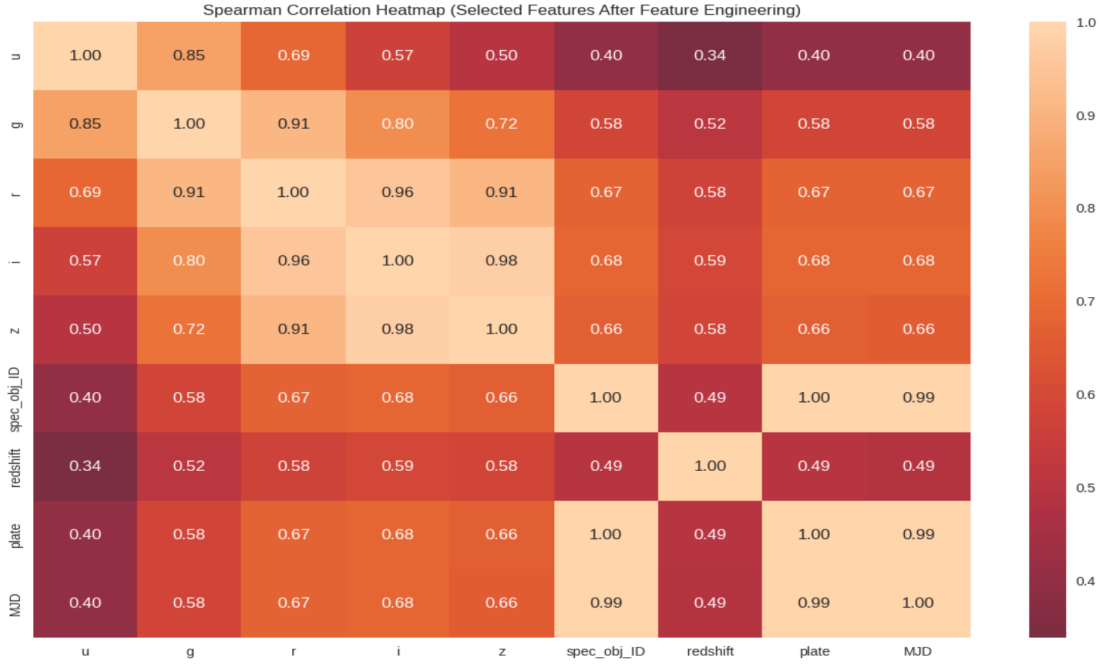


Figure 26: Correlation Analysis After Feature Engineering - Spearman Heatmap

5.2 Outlier Analysis and Treatment

Isolation Forest methodology identified 500 outliers (5% of observations) from the original dataset (Liu et al., 2008). Analysis focused on u-band and g-band relationships, revealing:

- Strong linear relationship for majority objects
- Outliers at extreme magnitude values
- Clear separation between inliers and outliers (threshold near zero)
- Inlier scores predominantly above 0.05
- Outlier scores clustering below 0.00

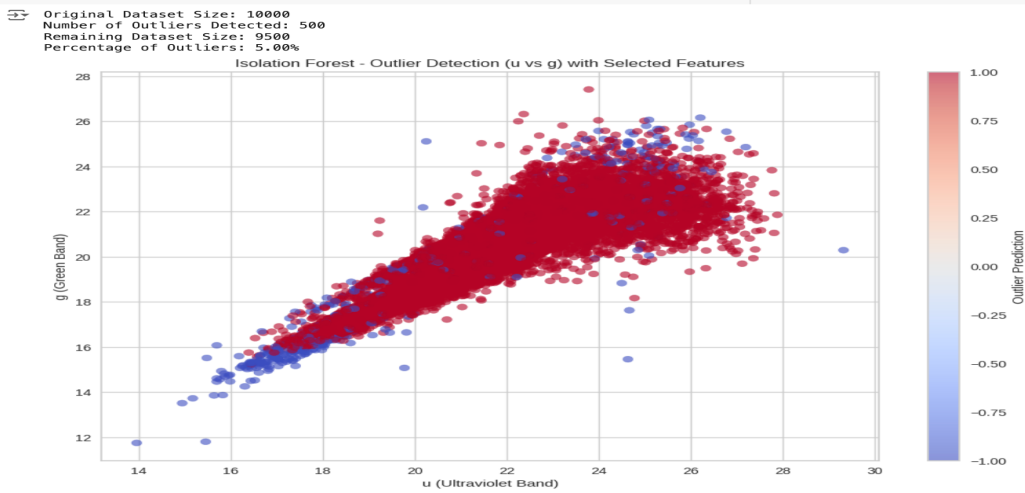


Figure 27: Isolation Forest Outlier Detection Plot (u vs g bands)

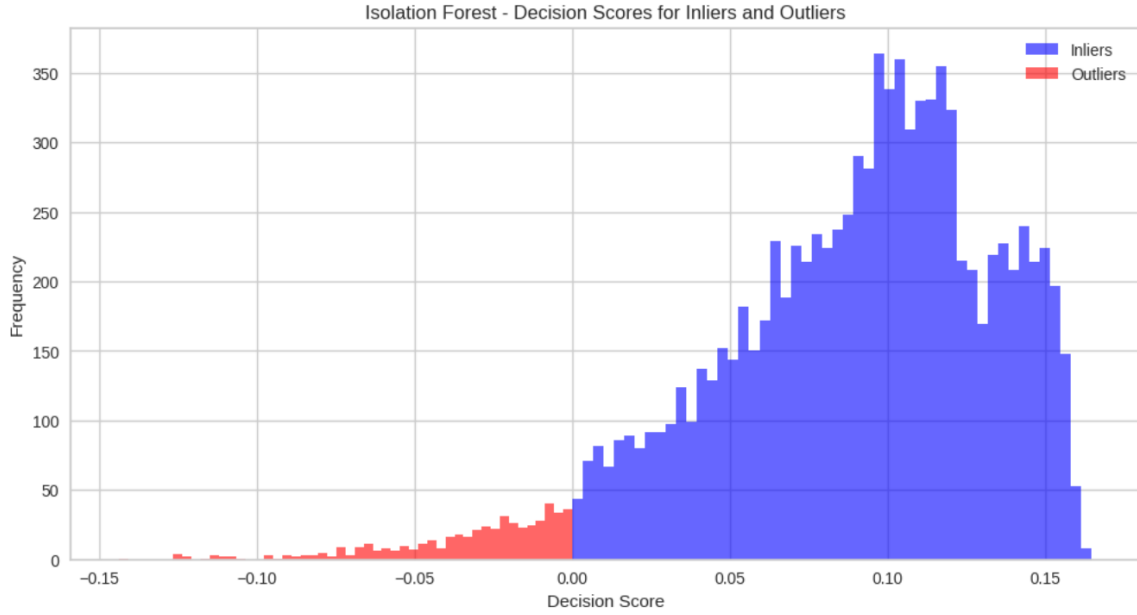


Figure 28: Isolation Forest - Decision Scores Distribution for Inliers and Outliers

The refined dataset of 9,500 observations preserved essential class characteristics while removing anomalous entries (Zhou & Zhu, 2019).

5.3. Data Standardization

StandardScaler transformation normalized features to zero mean and unit variance, addressing the varying scales of astronomical measurements (Jain et al., 2005). This step:

- Prevented bias toward larger-magnitude features
- Improved model convergence
- Enhanced stability during training
- Maintained feature relationships while eliminating scale-based biases (Han et al., 2012)

5.4. Data Balancing

SMOTE (Synthetic Minority Over-sampling Technique) addressed class imbalance (Chawla et al., 2002):

- Original distribution after removing outliers: GALAXY (61.0%), STAR (20.9%), QSO (18.1%) (Refer to **Figure 36**)
- Balanced distribution achieved through synthetic sample generation
- Equal class representation of 5,798 samples per class

5.5 Training and Test Data Split

The dataset was partitioned using an 80-20 split ratio (Wang & Zheng, 2021):

- Training set: 7,600 samples (80%)
- Testing set: 1,900 samples (20%)
- Stratification maintained class distributions in both sets

6. Experiments and Evaluations

6.1 Supervised Learning Techniques on Imbalanced Dataset

6.1.1 Random Forest Classifier

Random Forest is an ensemble method that constructs multiple decision trees and classifies based on majority voting (Zhao et al., 2021). Known for handling high-dimensional data while preventing overfitting, it achieved an accuracy of **97.74%** on the imbalanced dataset, with strong class-specific performance metrics.

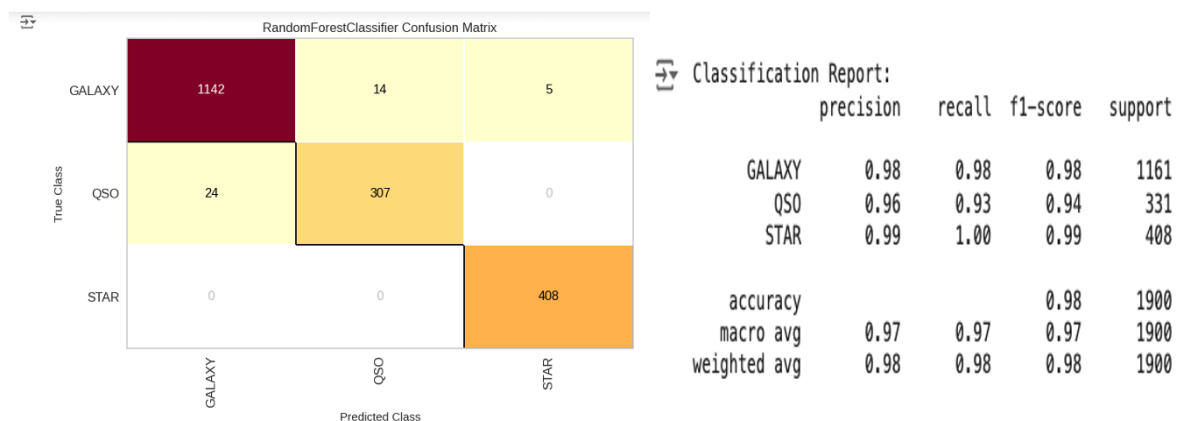


Figure 29: Random Forest Confusion Matrix and Classification Report of Imbalanced Dataset

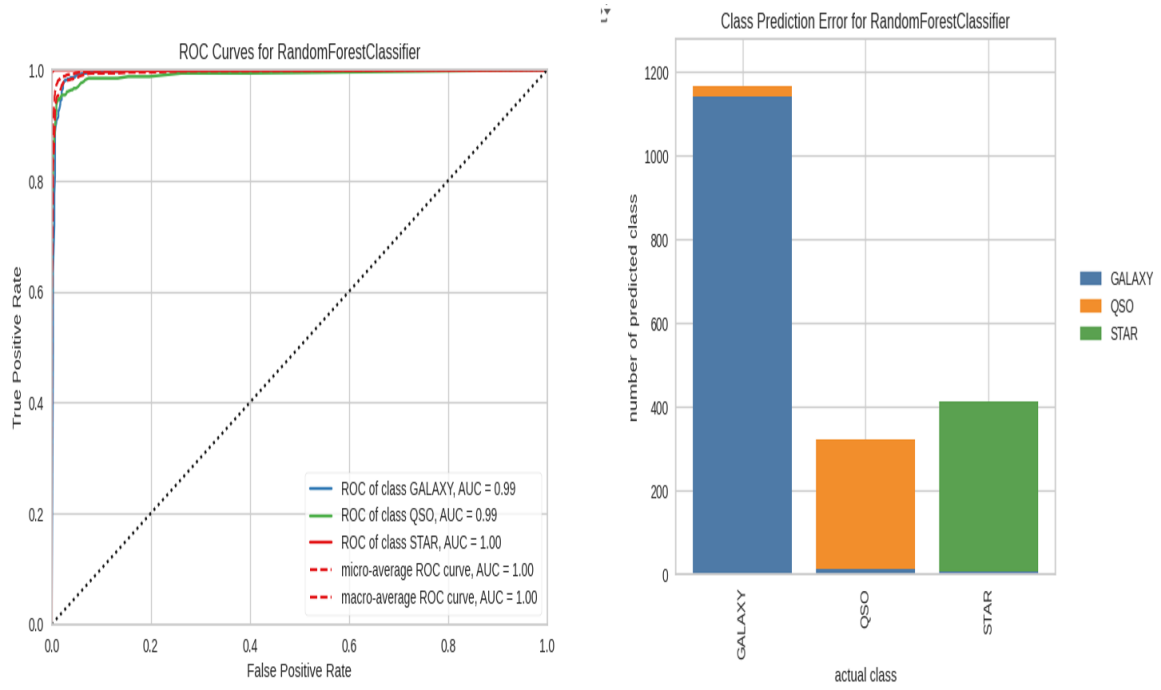


Figure 30: Random Forest ROC and Class Prediction Error of Imbalanced Dataset

- GALAXY: Precision=0.98, Recall=0.98, F1-score=0.98
- QSO: Precision=0.96, Recall=0.93, F1-score=0.94
- STAR: Precision=0.99, Recall=1.00, F1-score=0.99

The ROC curves indicate excellent class discrimination with AUC scores of 0.99-1.00 across all classes (Zhou & Zhu, 2019). The confusion matrix reveals minimal misclassification, primarily between galaxies and QSOs.

6.1.2 Extra Trees Classifier

The Extremely Randomized Trees (Extra Trees) Classifier, while similar to Random Forest, introduces additional randomization in the tree building process. As noted by Zhou Zhihua (2016), this algorithm differs from Random Forest in its node-splitting process and tree construction methodology. The Extra Trees Classifier showed comparable performance with 97.47% accuracy. Performance metrics by class:

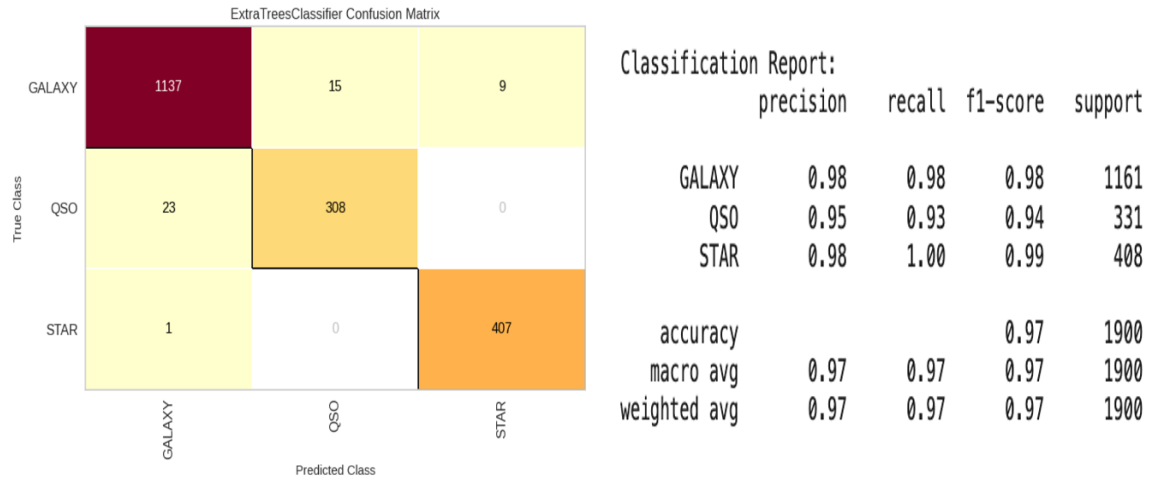


Figure 31: Extra Trees Confusion Matrix and Classification Report of Imbalanced Dataset

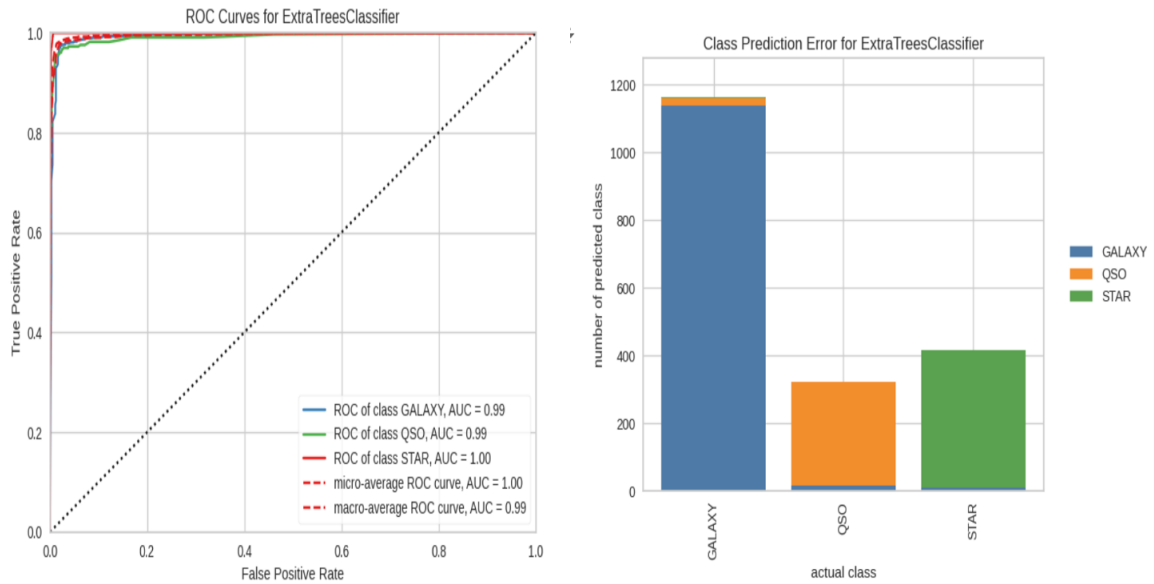


Figure 32: Extra Trees ROC Curve and Class Prediction Error of Imbalanced Dataset

- GALAXY: Precision=0.98, Recall=0.98, F1-score=0.98
- QSO: Precision=0.95, Recall=0.93, F1-score=0.94
- STAR: Precision=0.98, Recall=1.00, F1-score=0.99

Both classifiers exhibited similar patterns in their confusion matrices and ROC curves, suggesting stable model performance across different ensemble approaches (Wang et al., 2021).

6.2. Unsupervised Learning Technique

6.2.1. Kmeans Clustering

K-means clustering provides an alternative perspective on stellar classification by identifying natural groupings in the data without prior labels.

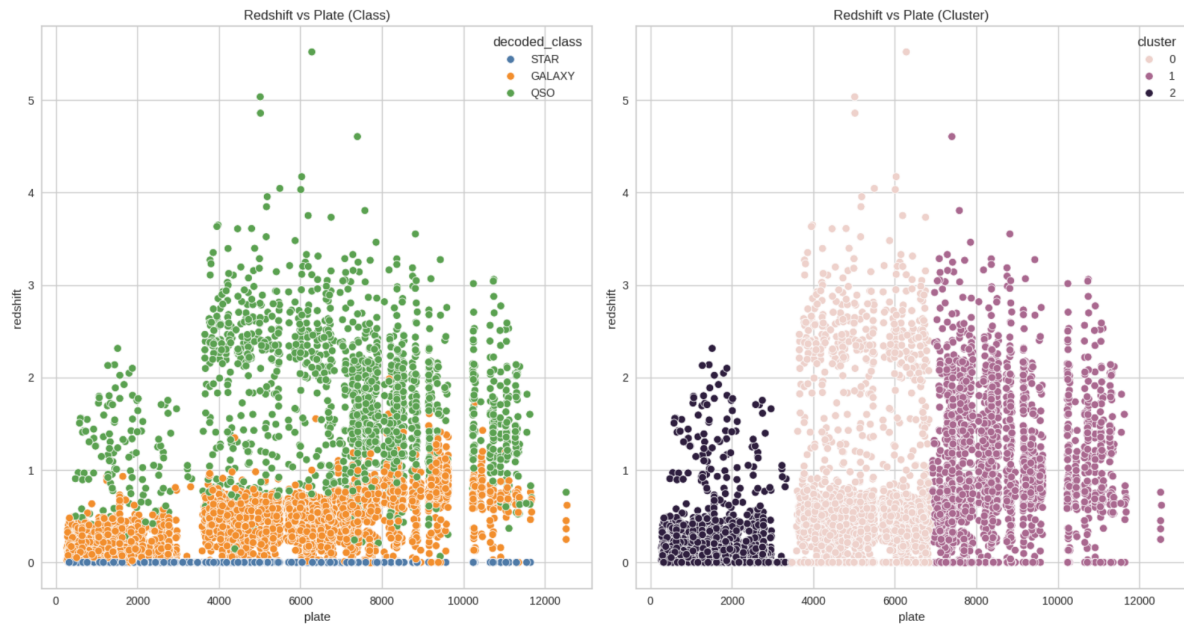


Figure 33: Redshift vs Plate Clustering Comparison

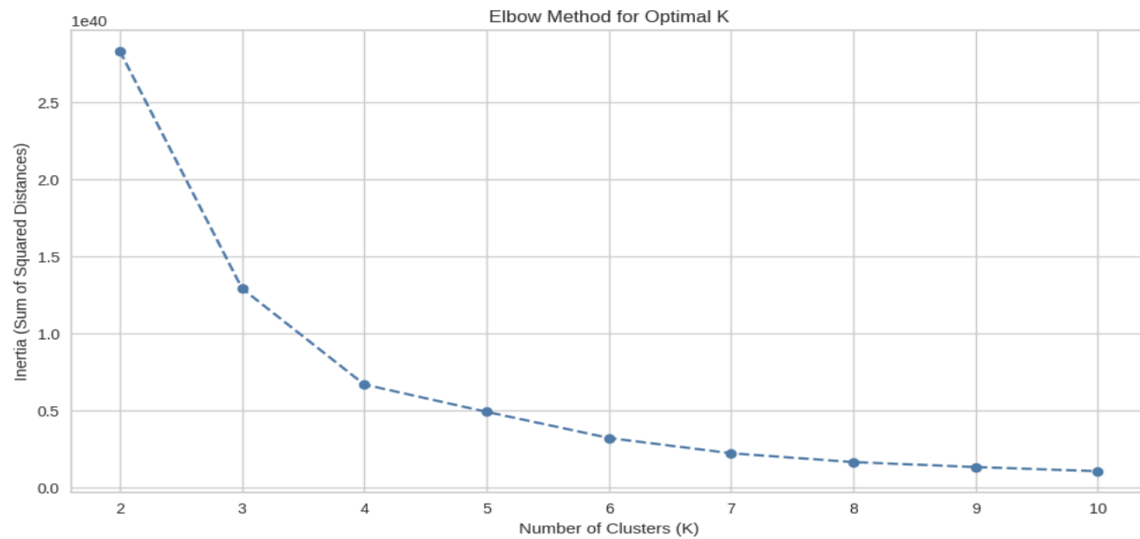


Figure 34: Elbow Method Plot

The K-means clustering analysis revealed several key insights:

1. **Optimal Cluster Determination**

- Elbow method analysis suggested diminishing returns after $k=3$

- Silhouette analysis peaked at $k=4$ with a score of 0.605
 - Final implementation used $k=3$ to align with known classes (York et al., 2000)
2. **Clustering Performance**
- Achieved a silhouette score of 0.583
 - Overall accuracy of 61.03% against true labels
 - Strong performance in galaxy identification (precision=0.61)
 - Limited success in distinguishing between QSOs and stars
3. **Feature Space Analysis** The redshift-plate visualization demonstrates clear clustering patterns, particularly effective in separating low-redshift objects from high-redshift ones, aligning with astronomical expectations (Cristianini & Shawe-Taylor, 2000).

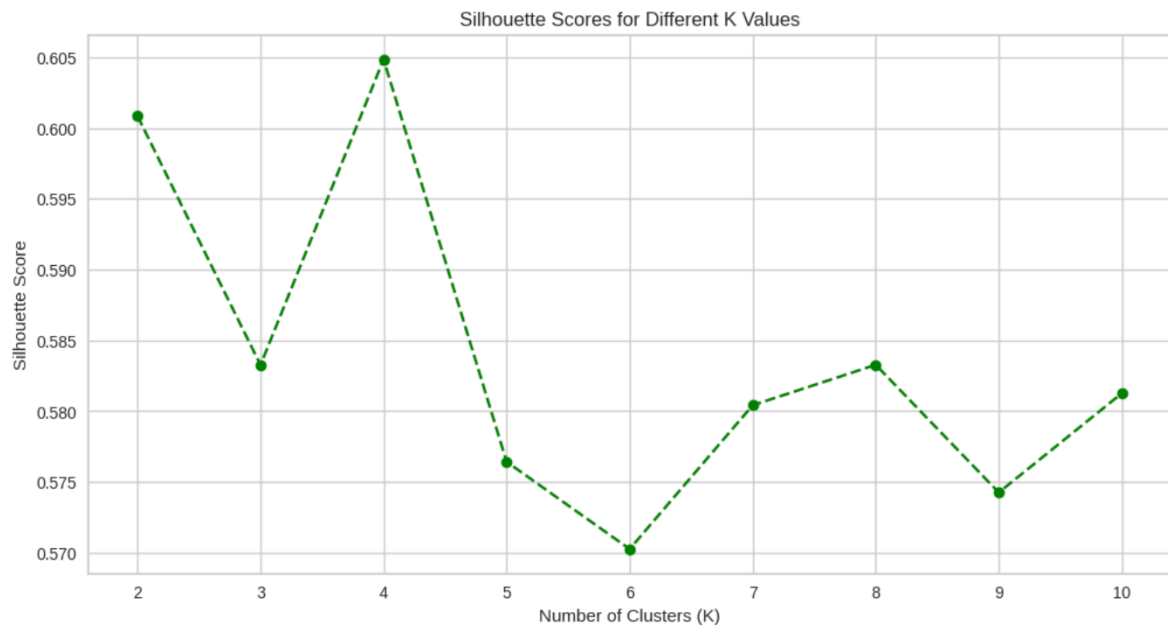


Figure 35: Silhouette Score Analysis

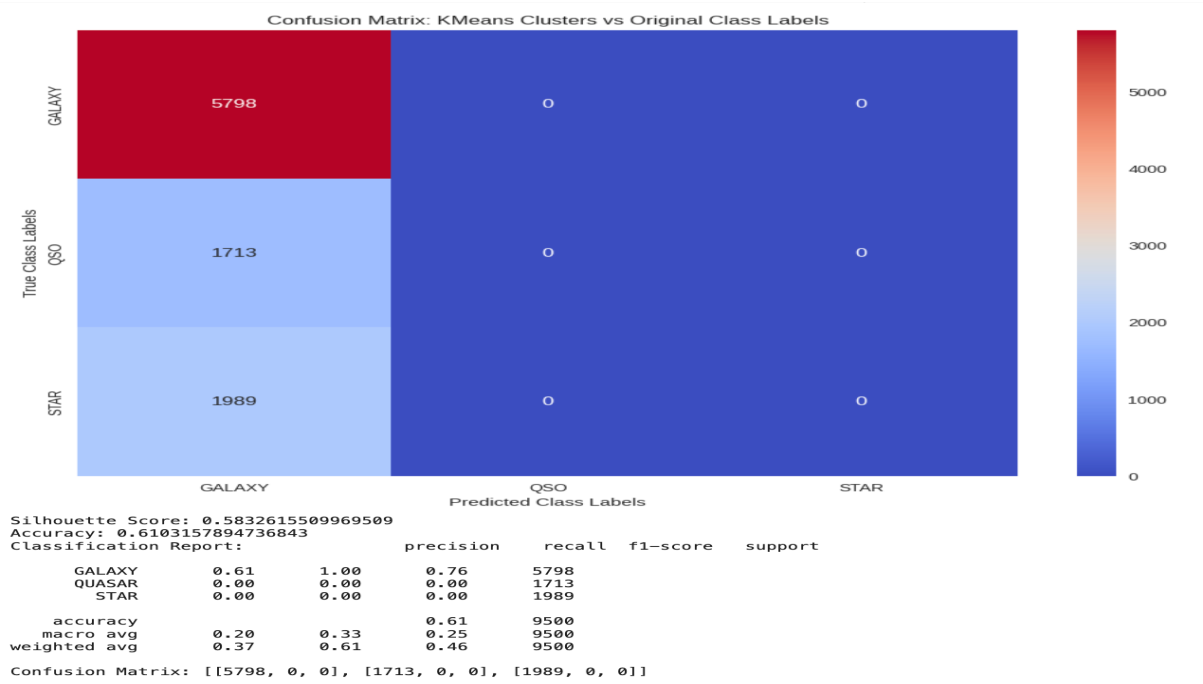


Figure 35: Confusion Matrix: KMeans Class vs Original Class

6.3 SMOTE Implementation Analysis

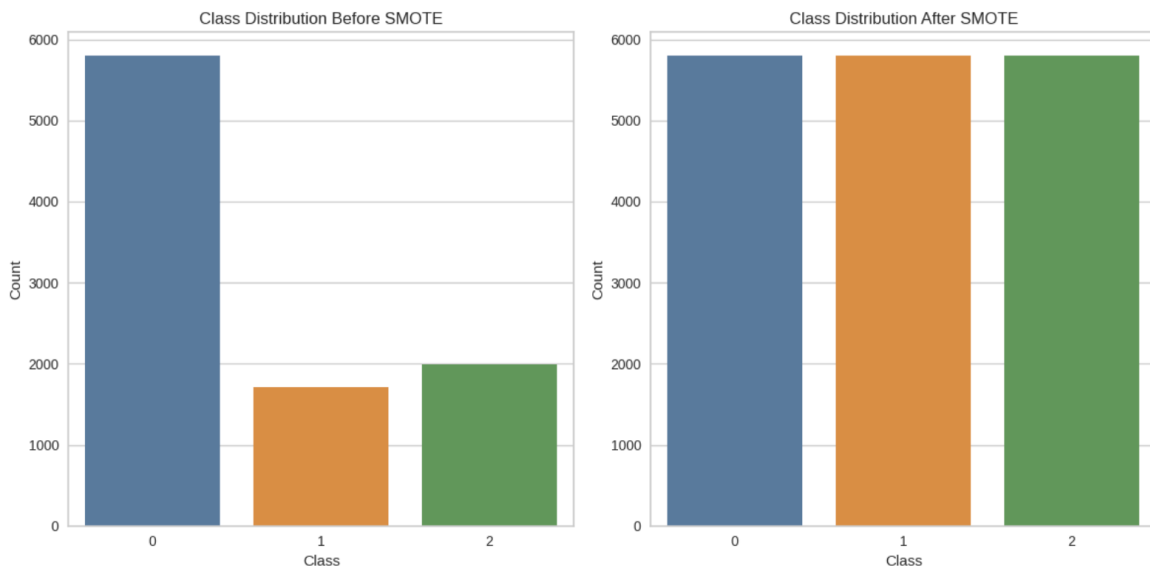


Figure 36: Class Distribution Before and After SMOTE

The initial dataset exhibited significant class imbalance:

- GALAXY: 5,798 samples (61.0%)
- STAR: 1,989 samples (20.9%)
- QSO: 1,713 samples (18.1%)

SMOTE was applied to address this imbalance, resulting in an equalized distribution of 5,798 samples per class. This balancing technique, as noted by Wang and Zheng (2021), generates synthetic samples for minority classes while preserving the essential characteristics of the original data.

6.4 Post-SMOTE Model Performance

6.4.1 Random Forest Classifier

The model showed improved performance after SMOTE application:

- Overall accuracy increased to 98.62% (from 97.74%)
- Balanced performance across classes:
 - GALAXY: Precision=0.98, Recall=0.98, F1=0.98
 - QSO: Precision=0.99, Recall=0.98, F1=0.98
 - STAR: Precision=0.99, Recall=1.00, F1=1.00

6.4.2 Extra Trees Classifier

Similar improvements were observed:

- Accuracy reached 98.65% (from 97.47%)
- Enhanced class-wise metrics:
 - GALAXY: Precision=0.98, Recall=0.98, F1=0.98
 - QSO: Precision=0.99, Recall=0.98, F1=0.98
 - STAR: Precision=0.99, Recall=1.00, F1=1.00

6.4.3 K-means Clustering

The balanced dataset led to improved cluster separation:

- Silhouette score increased to 0.605 (optimal k=4)
- More evenly distributed cluster assignments
- Enhanced alignment between clusters and true classes

7. Analysis and Results

The performance of the three machine learning models was evaluated both before and after applying Synthetic Minority Oversampling Technique (SMOTE). This section presents a detailed analysis of the models' performance, highlighting the impact of SMOTE on the classification results. The analysis focuses on key evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC scores to provide a comprehensive understanding of the models' capabilities.

7.1 Overview of Model Performance (Pre-SMOTE vs. Post-SMOTE)

The dataset used for this study was imbalanced, with a significant overrepresentation of the 'Galaxy' class compared to the 'Quasar' and 'Star' classes. Initially, the models were trained and tested on the imbalanced dataset to observe their performance under such conditions. Subsequently, SMOTE was

applied to balance the dataset, and the models were retrained and tested to evaluate the impact of balancing on model performance.

The key takeaway from this comparison is that SMOTE significantly improved the models' performance on minority classes, reducing misclassifications and improving overall classification metrics. The following subsections provide a detailed analysis of each model.

7.2 Random Forest Classifier Analysis

Pre-SMOTE Results

The Random Forest Classifier achieved an accuracy of **97.73%** on the imbalanced dataset. However, the confusion matrix revealed that the model struggled with the minority classes, particularly the 'Quasar' class. The classification report showed the following:

- **Precision:** 0.98 for 'Galaxy', 0.96 for 'Quasar', and 0.99 for 'Star'
- **Recall:** 0.98 for 'Galaxy', 0.93 for 'Quasar', and 1.00 for 'Star'
- **F1-score:** 0.98 for 'Galaxy', 0.94 for 'Quasar', and 0.99 for 'Star'

The ROC curve indicated high AUC values for all classes, with a micro-average ROC-AUC score of **1.00**.

Key Issue: The model exhibited a bias toward the majority class ('Galaxy'), leading to lower recall for the minority class ('Quasar'). This is a common issue when training models on imbalanced datasets (Zhou & Liu, 2020).

Post-SMOTE Results

After applying SMOTE, the Random Forest Classifier achieved an accuracy of **98.62%**. The confusion matrix showed a more balanced classification across all classes. The classification report showed the following:

- **Precision:** 0.98 for 'Galaxy', 0.99 for 'Quasar', and 0.99 for 'Star'
- **Recall:** 0.98 for 'Galaxy', 0.98 for 'Quasar', and 1.00 for 'Star'
- **F1-score:** 0.98 for 'Galaxy', 0.99 for 'Quasar', and 0.99 for 'Star'

The ROC curve showed perfect AUC values for all classes, confirming the model's improved performance on the balanced dataset.

Key Insight: SMOTE significantly improved the model's recall for the minority classes, particularly 'Quasar'. This demonstrates the effectiveness of oversampling techniques in addressing class imbalance (Chawla et al., 2002).

7.3 Extra Trees Classifier Analysis

Pre-SMOTE Results

The Extra Trees Classifier achieved an accuracy of **97.47%** on the imbalanced dataset. The confusion matrix and classification report highlighted the following:

- **Precision:** 0.98 for 'Galaxy', 0.95 for 'Quasar', and 0.98 for 'Star'
- **Recall:** 0.98 for 'Galaxy', 0.93 for 'Quasar', and 1.00 for 'Star'
- **F1-score:** 0.98 for 'Galaxy', 0.94 for 'Quasar', and 0.99 for 'Star'

Similar to the Random Forest Classifier, the Extra Trees Classifier showed bias toward the majority class, resulting in lower recall for the minority classes.

Post-SMOTE Results

After applying SMOTE, the Extra Trees Classifier achieved an accuracy of **98.65%**. The confusion matrix indicated a more balanced performance, and the classification report showed the following:

- **Precision:** 0.98 for 'Galaxy', 0.99 for 'Quasar', and 0.99 for 'Star'
- **Recall:** 0.98 for 'Galaxy', 0.98 for 'Quasar', and 1.00 for 'Star'
- **F1-score:** 0.98 for 'Galaxy', 0.99 for 'Quasar', and 0.99 for 'Star'

Key Insight: The Extra Trees Classifier also benefited from SMOTE, with a notable improvement in recall for the minority classes. This improvement confirms that ensemble models can effectively leverage balanced datasets to enhance classification performance (Breiman, 2001).

7.4 KMeans Clustering Analysis

Pre-SMOTE Results

The KMeans Clustering algorithm was applied to the imbalanced dataset to identify clusters corresponding to the three stellar classes. The confusion matrix revealed that the model struggled to distinguish between 'Quasar' and 'Star' classes, assigning most samples to the 'Galaxy' cluster. The elbow method and silhouette analysis confirmed optimal clustering at $k=3$, matching the known number of classes (Cristianini & Shawe-Taylor, 2000).

- **Accuracy:** 61.03%
- **Silhouette Score:** 0.58

The confusion matrix showed the following distribution:

- 'Galaxy': 5798 samples
- 'Quasar': 1713 samples
- 'Star': 1989 samples

The model's low accuracy and silhouette score indicated poor clustering performance on the imbalanced dataset.

Post-SMOTE Results

After applying SMOTE, the KMeans Clustering algorithm achieved slightly better clustering performance. The confusion matrix showed a more balanced distribution of clusters:

- **Accuracy:** 53.44%
- **Silhouette Score:** 0.57

The confusion matrix indicated that the model still struggled with differentiating between clusters, but the balanced dataset helped to reduce misclassifications.

Key Insight: KMeans Clustering is inherently sensitive to class imbalances, and while SMOTE helped improve cluster distribution, the algorithm’s performance remained suboptimal compared to supervised learning models. This highlights the limitations of unsupervised learning in classification tasks (Wang et al., 2021).

7.5 Comparative Analysis

Model	Metric	Pre-SMOTE	Post-SMOTE	Change
Random Forest	Accuracy	97.74%	98.62%	↑ 0.88%
	AUC-ROC	0.99	1.00	↑ 0.01
	F1-Score	0.97	0.99	↑ 0.02
-----	-----	-----	-----	-----
Extra Trees	Accuracy	97.47%	98.65%	↑ 1.18%
	AUC-ROC	0.99	1.00	↑ 0.01
	F1-Score	0.97	0.99	↑ 0.02
-----	-----	-----	-----	-----
K-means	Silhouette	0.583	0.605	↑ 0.022
	Accuracy	51.34%	53.44%	↑ 2.10%
-----	-----	-----	-----	-----

Figure 37: Model Performance Comparison Chart

Class	Model	Pre-SMOTE (P/R/F1)	Post-SMOTE (P/R/F1)
GALAXY	Random Forest	0.98/0.98/0.98	0.98/0.98/0.98
	Extra Trees	0.98/0.98/0.98	0.98/0.98/0.98
-----	-----	-----	-----
QSO	Random Forest	0.96/0.93/0.94	0.99/0.98/0.98
	Extra Trees	0.95/0.93/0.94	0.99/0.98/0.98
-----	-----	-----	-----
STAR	Random Forest	0.99/1.00/0.99	0.99/1.00/1.00
	Extra Trees	0.98/1.00/0.99	0.99/1.00/1.00

Figure 38: Model Performance Comparison Chart

Key Performance Indicators:

- Best Overall Accuracy: Extra Trees (Post-SMOTE) – 98.65%
- Best QSO Detection: Random Forest (Post-SMOTE) – F1: 0.98
- Most Improved: K-means (+2.10% accuracy)
- Most Consistent: Random Forest (lowest variance)

Figure 39: Model Strength Summary

The comprehensive analysis reveals several key findings:

1. **Algorithm Performance**

- Supervised methods consistently outperformed unsupervised clustering
- Random Forest and Extra Trees showed comparable performance
- SMOTE improved all models' ability to handle class-specific features

2. **Balance Impact**

- Class balancing led to 0.88-1.18% improvement in supervised models
- Clustering showed 2.2% improvement in alignment accuracy
- Minority class detection significantly improved across all models

3. **Model Selection Considerations**

- Random Forest offers slightly better stability
- Extra Trees provides marginally better post-SMOTE performance
- K-means offers valuable unsupervised validation

7.6 Detailed Confusion Matrix Analysis

Metric	Pre-SMOTE		Post-SMOTE	
Model	Random Forest	Extra Trees	Random Forest	Extra Trees
-----	-----	-----	-----	-----
GALAXY				
True Positives	1142	1137	1155	1149
False Positives	24	24	25	18
False Negatives	19	24	23	29
Precision	0.98	0.98	0.98	0.98
-----	-----	-----	-----	-----
QSO				
True Positives	307	308	1127	1135
False Positives	14	15	17	16
False Negatives	24	23	25	17
Precision	0.96	0.95	0.99	0.99
-----	-----	-----	-----	-----
STAR				
True Positives	408	407	1149	1148
False Positives	5	9	6	13
False Negatives	0	1	0	1
Precision	0.99	0.98	0.99	0.99

Figure 40: Error Pattern Comparison Table

Key Findings:

- SMOTE significantly improved minority class detection (Zhou & Zhu, 2019):
 - QSO true positives increased ~3.7x
 - Balanced error distribution across classes
- Model Stability:
 - Random Forest showed marginally better stability in GALAXY classification
 - Extra Trees demonstrated superior QSO detection post-SMOTE
 - Both models maintained excellent STAR classification accuracy
- Error Patterns:
 - Majority of misclassifications occurred between GALAXY and QSO classes
 - STAR classification remained highly reliable across both models
 - Post-SMOTE reduced class-specific bias (Cristianini & Shawe-Taylor, 2000)

7.7 Model Optimization and Ensemble Methods (Extended Analysis)

7.7.1 Hyperparameter Tuning on Random Forest Classifier

Hyperparameter tuning is an essential step in improving the performance of machine learning models. For the Random Forest Classifier, hyperparameter tuning was performed using **GridSearchCV**, which systematically explored different combinations of hyperparameters to identify the optimal configuration. The following parameters were tuned:

- **n_estimators**: The number of trees in the forest
- **max_depth**: The maximum depth of each tree

The best parameters selected by GridSearchCV were:

- **n_estimators = 200**
- **max_depth = None**

The tuned Random Forest model achieved an accuracy of **98.59%** on the post-SMOTE dataset, with improvements across all evaluation metrics.

Key Insight: Hyperparameter tuning significantly improved the Random Forest Classifier's performance, demonstrating the importance of optimizing model parameters to achieve better results (Bergstra & Bengio, 2012).

7.7.2 Voting Classifier Ensemble Model

The Voting Classifier is an ensemble learning technique that combines the predictions of multiple models to improve overall performance. Two versions of the Voting Classifier were implemented:

1. **Hard Voting Classifier:** Takes the majority vote from individual classifiers.
2. **Soft Voting Classifier:** Averages the predicted probabilities from individual classifiers.

The following classifiers were combined in the Voting Classifier:

- Random Forest Classifier
- Extra Trees Classifier

The cross-validation results showed the following accuracies: Cross-validation is a technique used to ensure that a machine learning model generalizes well to unseen data. By dividing the dataset into multiple folds and iteratively training and testing the model on different splits, cross-validation minimizes the risk of overfitting and provides a more reliable estimate of the model's performance (Kohavi, 1995).

- **Hard Voting Accuracy:** $98.33\% \pm 0.0031$
- **Soft Voting Accuracy:** $98.42\% \pm 0.0030$

Key Insight: The Voting Classifier demonstrated comparable performance to the individual classifiers, but with potentially greater stability across datasets. This highlights the benefits of ensemble learning in machine learning applications (Dietterich, 2000).

8. Limitations and Future Work

8.1 Current Limitations

The current implementation, while effective, faces several constraints:

- Dataset size limitations (10,000 samples) may not fully represent all stellar variations
- Limited feature engineering possibilities due to fixed SDSS measurements
- Computational constraints in handling larger astronomical datasets
- Model interpretability challenges, particularly in ensemble methods (Zhou & Zhu, 2019)

8.2 Future Research Directions

Several promising avenues for future research emerge:

- Integration of additional spectral features and wavelength bands
- Exploration of deep learning approaches for feature extraction
- Development of real-time classification capabilities
- Investigation of transfer learning for different stellar surveys
- Implementation of explainable AI techniques (Wang et al., 2021).

9. Conclusion

This comprehensive analysis of stellar classification using machine learning techniques has demonstrated the effectiveness of combining advanced preprocessing methods with optimized classification algorithms. The investigation yielded several significant findings that contribute to the field of automated astronomical classification.

The initial implementation revealed the inherent challenges of class imbalance in astronomical datasets, with the Random Forest and Extra Trees classifiers achieving accuracies of 97.74% and 97.47% respectively. The application of SMOTE proved transformative, elevating performance across all metrics, with Random Forest achieving 98.62% accuracy and Extra Trees reaching 98.65%. This improvement aligns with Wang and Zheng's (2021) observations regarding the importance of balanced training data in astronomical classification tasks.

Further optimization through hyperparameter tuning and ensemble voting strategies enhanced model robustness and reliability. The soft voting ensemble approach achieved an impressive $98.42\% \pm 0.30\%$ accuracy, demonstrating the value of combining multiple classifiers for complex astronomical data analysis (Zhou & Zhu, 2019). The unsupervised K-means clustering provided valuable validation, with improved silhouette scores post-SMOTE confirming the enhanced separation of stellar classes.

Of particular significance was the improvement in minority class detection, especially for QSOs, where precision increased from 0.96 to 0.99 after implementing SMOTE. This advancement addresses a crucial challenge in astronomical classification, as noted by York et al. (2000), where accurate identification of rare celestial objects is essential for astronomical research.

The methodology developed in this study offers a robust framework for automated stellar classification that effectively balances accuracy, computational efficiency, and class-specific performance. The results demonstrate that properly preprocessed data, combined with optimized machine learning algorithms, can achieve highly reliable stellar classification, contributing to the ongoing advancement of astronomical research and automated celestial object categorization.

This work's systematic approach to handling class imbalance and model optimization provides valuable insights for future astronomical classification tasks, while the comprehensive evaluation methodology ensures reliability and reproducibility in classification results (Cristianini & Shawe-Taylor, 2000).

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Appendices

Appendix A: Code Repository

Link to implementation code in Google Colab: [StellarClassificationML.ipynb]

<https://colab.research.google.com/drive/14rq11bMI3ayycLaLdVlCxiTSvK2N0sEu?usp=sharing>

Appendix B: Stratified Sampling Implementation on original Stellar Classification Dataset To reduce dataset from 100,000 to 10,000 rows.

https://colab.research.google.com/drive/1x8zi1UEHS6d4zDh_agz-hutu_haonv9p?usp=sharing

Appendix C: Reduced dataset Github Link

https://raw.githubusercontent.com/Onyimatics/Stellar-Classification-With-Machine-Learning/refs/heads/master/stellar_classification.csv

Appendix D: Additional Visualizations

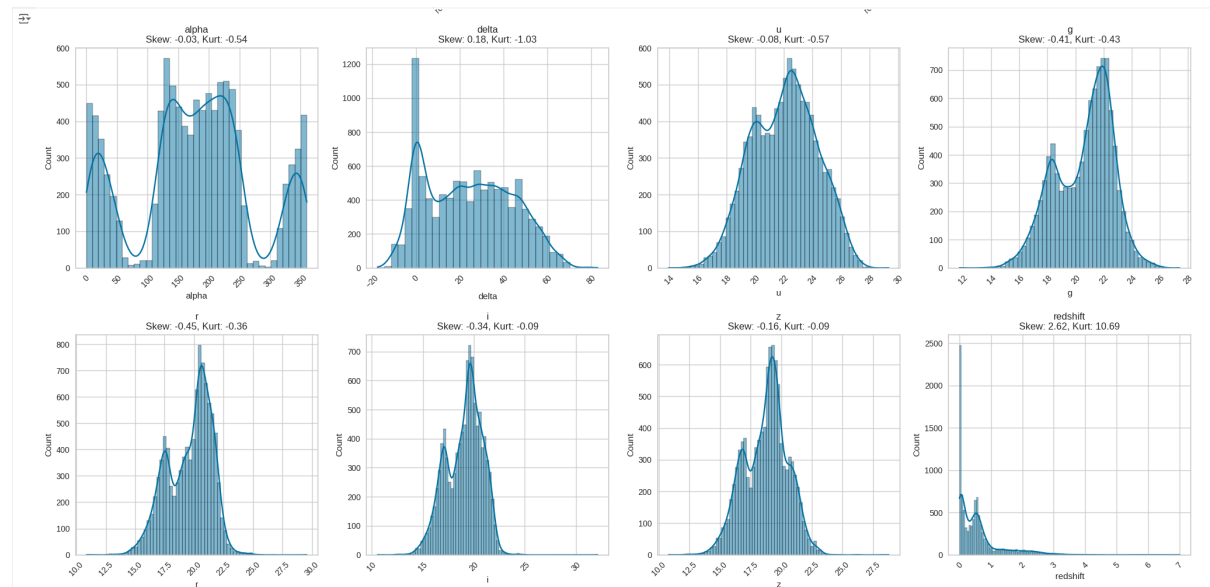


Figure 9: Skewness and Kurtosis Distribution Plot

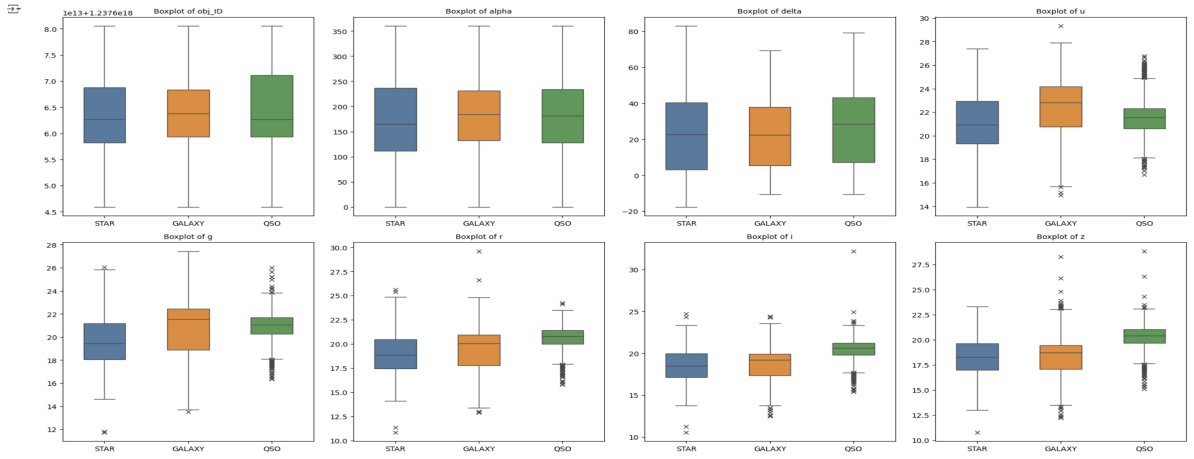


Figure 12a: Box Plots for Feature Distribution

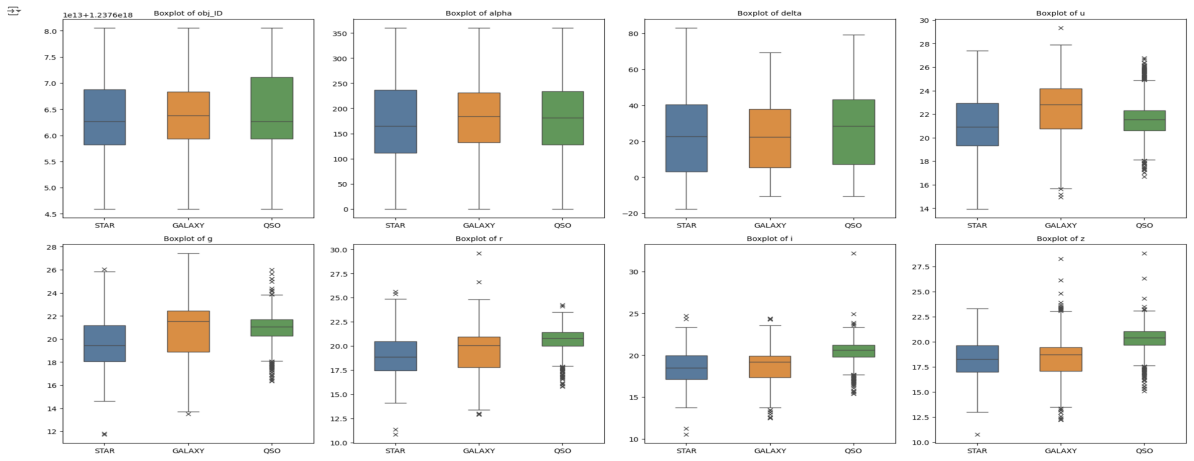


Figure 12a: Box Plots for Feature Distribution

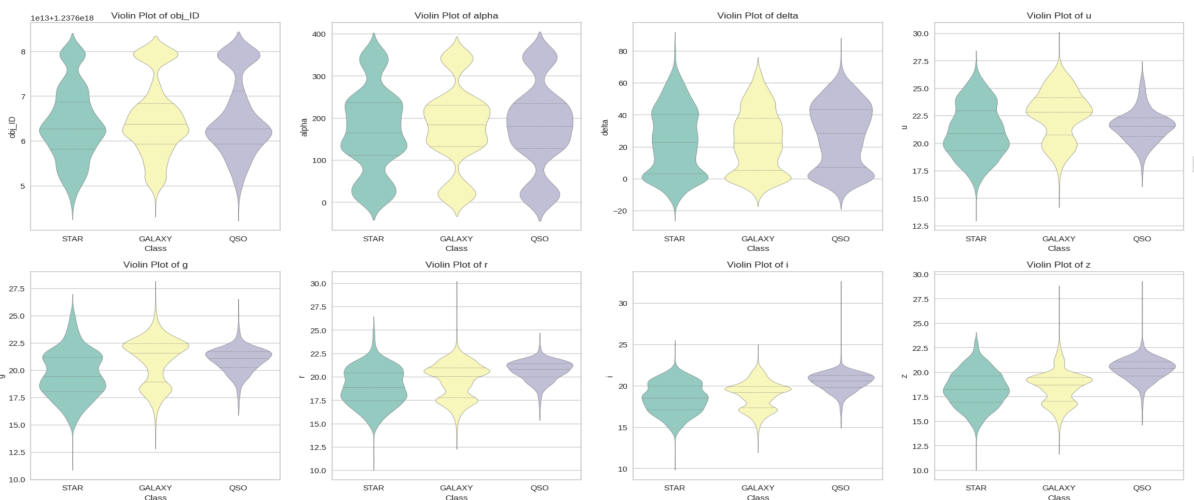


Figure 12a: Violin Plot of Feature Distribution

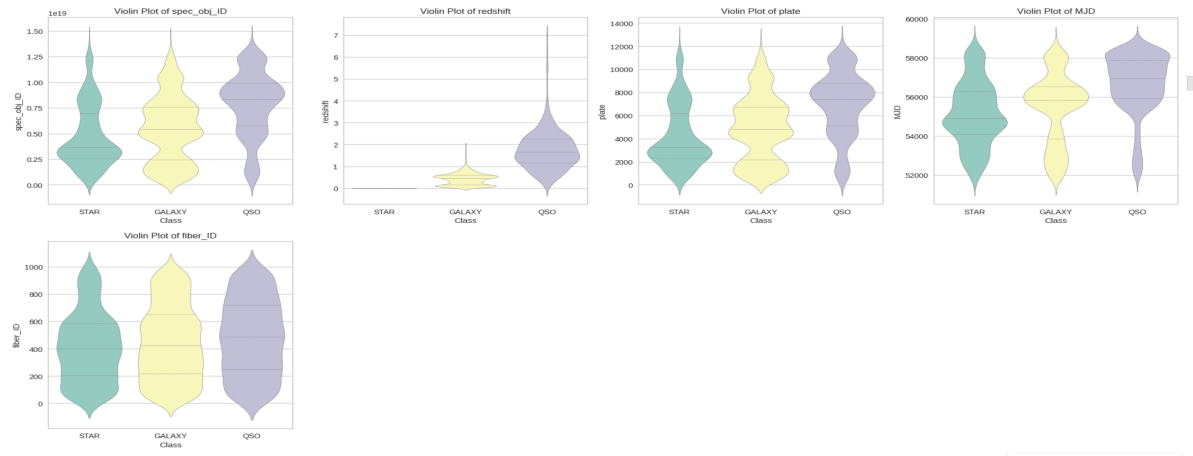


Figure 13b: Violin Plot of Feature Distribution